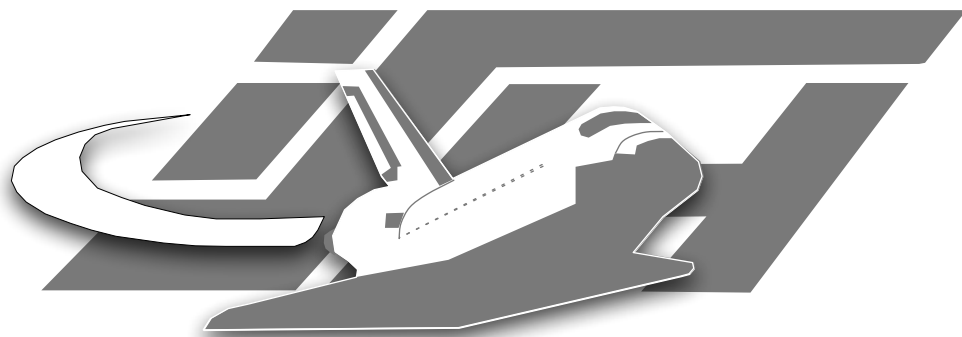


The costs and benefits of Honeynets

Maximillian Dornseif with Sascha A. May
Laboratory for Dependable Distributed Systems
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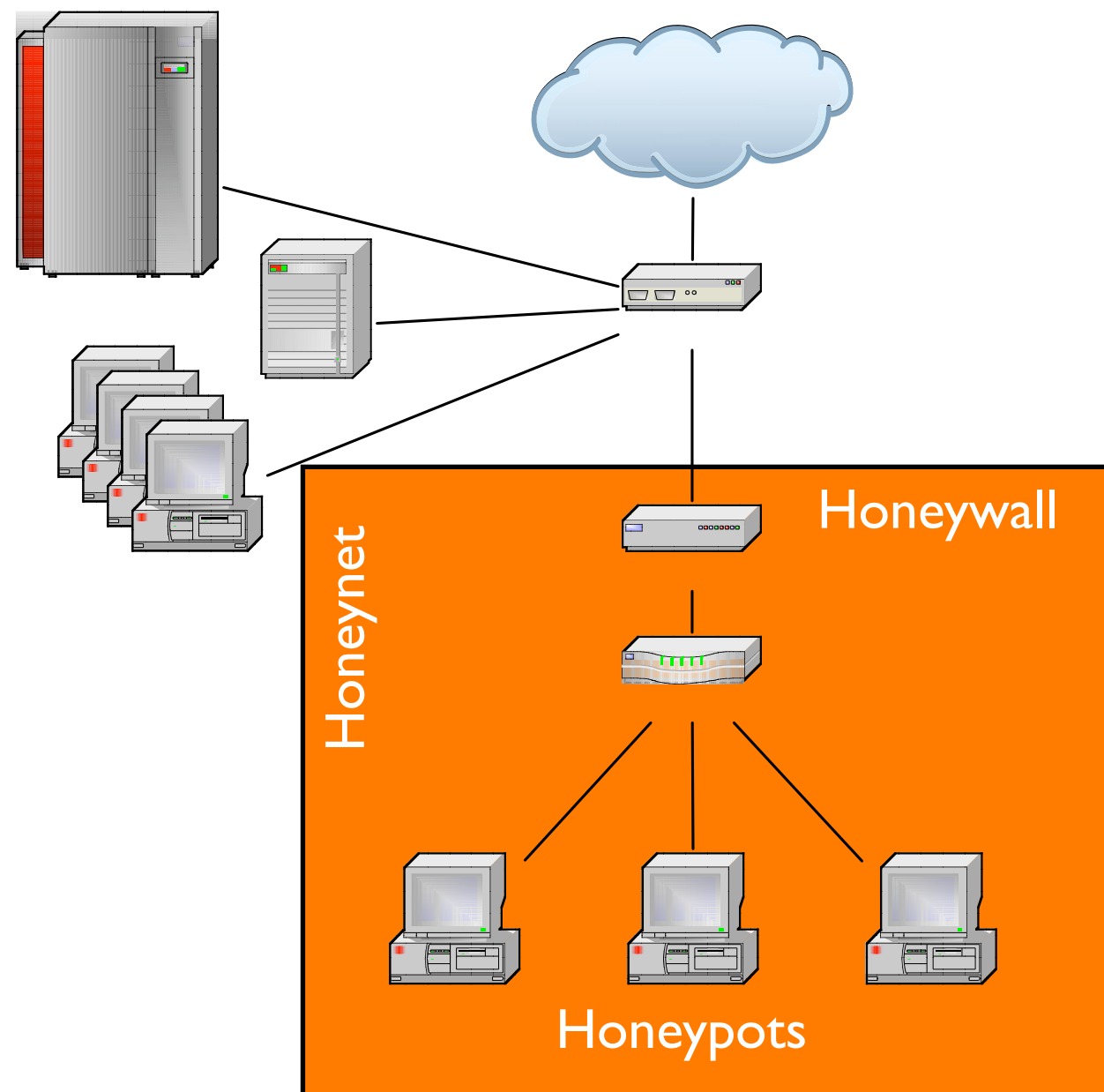
at

The 3. Workshop on Economics of Information Security
(WEIS 2004)

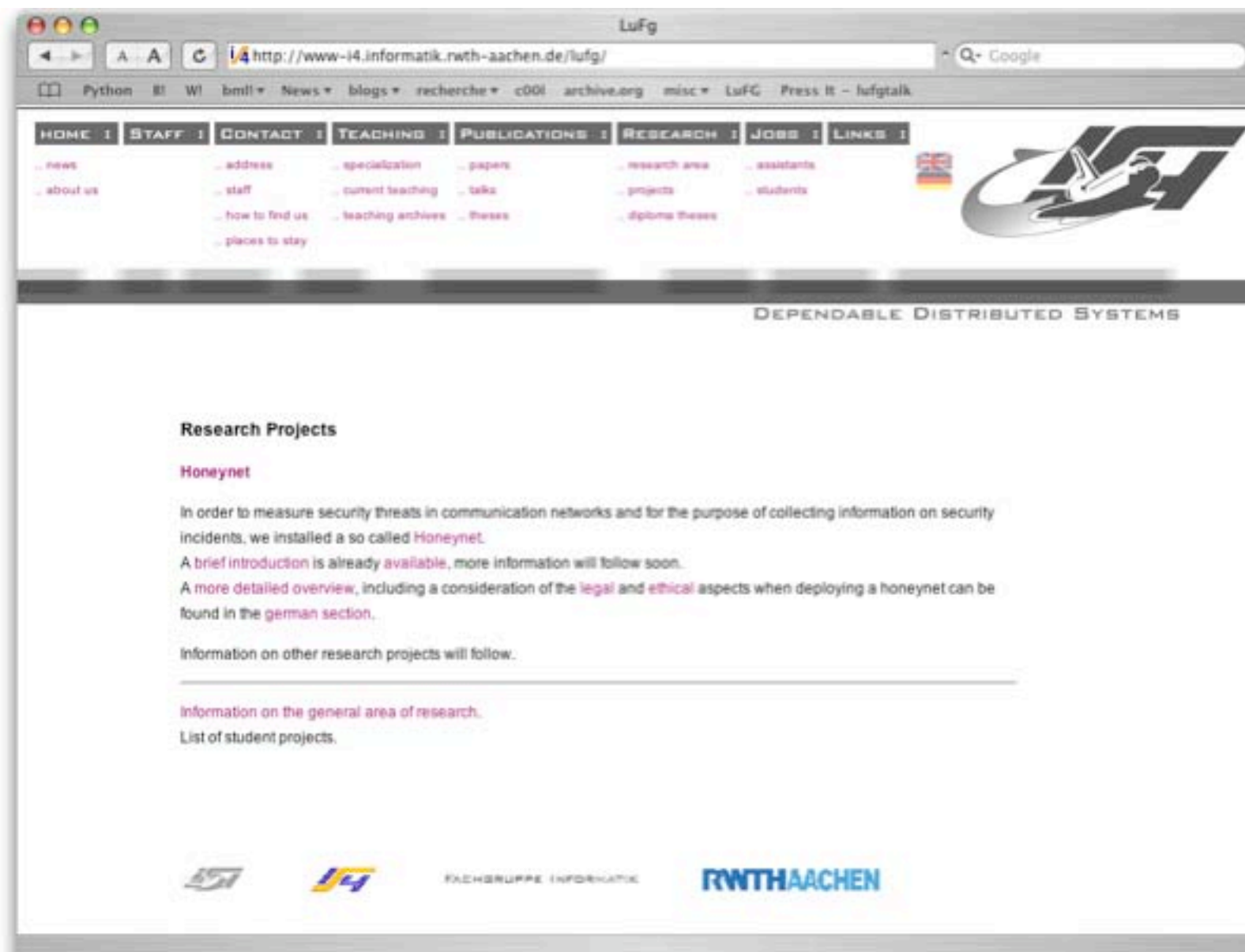


RWTHAACHEN
RHEINISCH-WESTFÄLISCHE TECHNISCHE HOCHSCHULE AACHEN

Honeynets



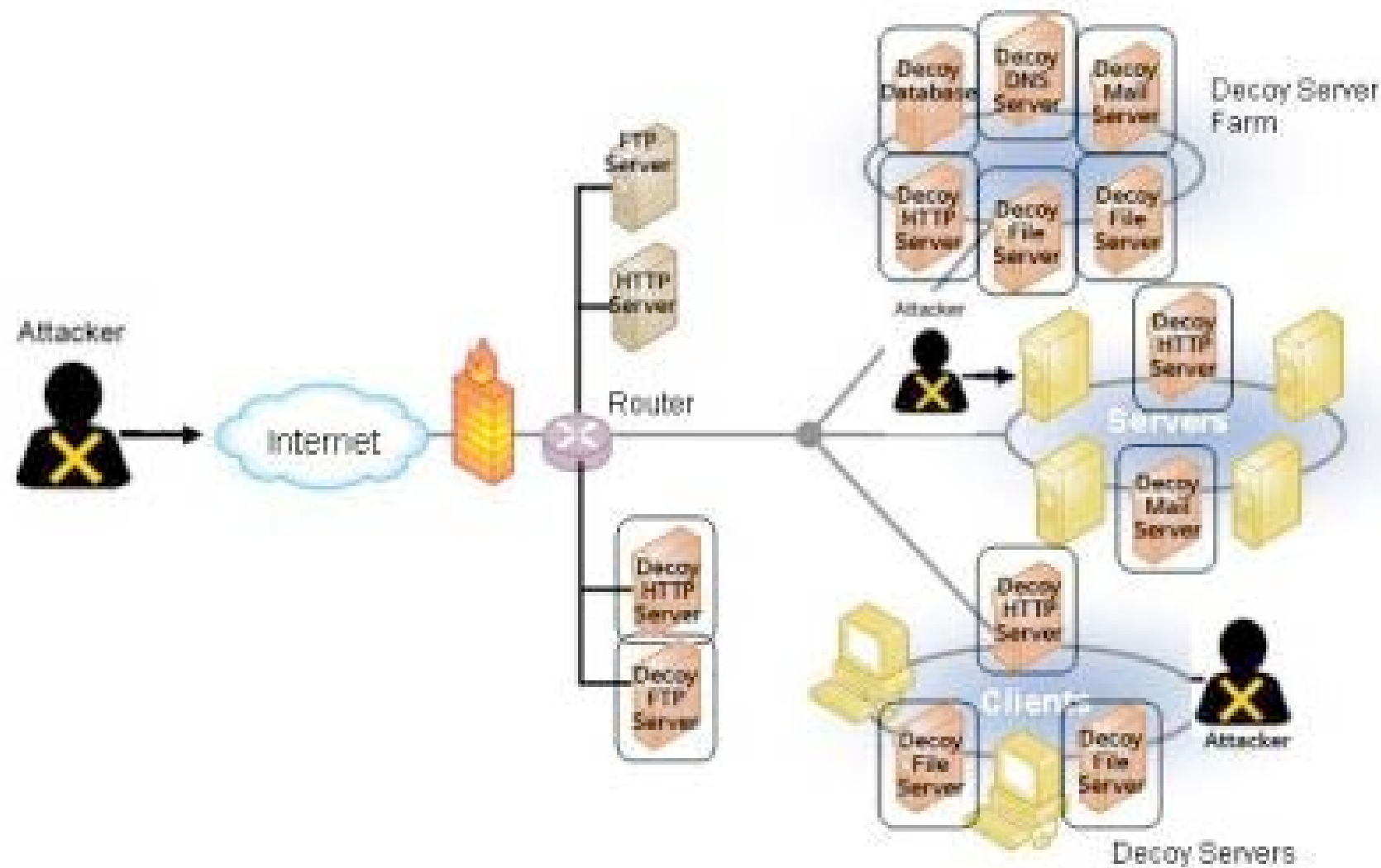
Motivation



Possible Benefits of Honeynets

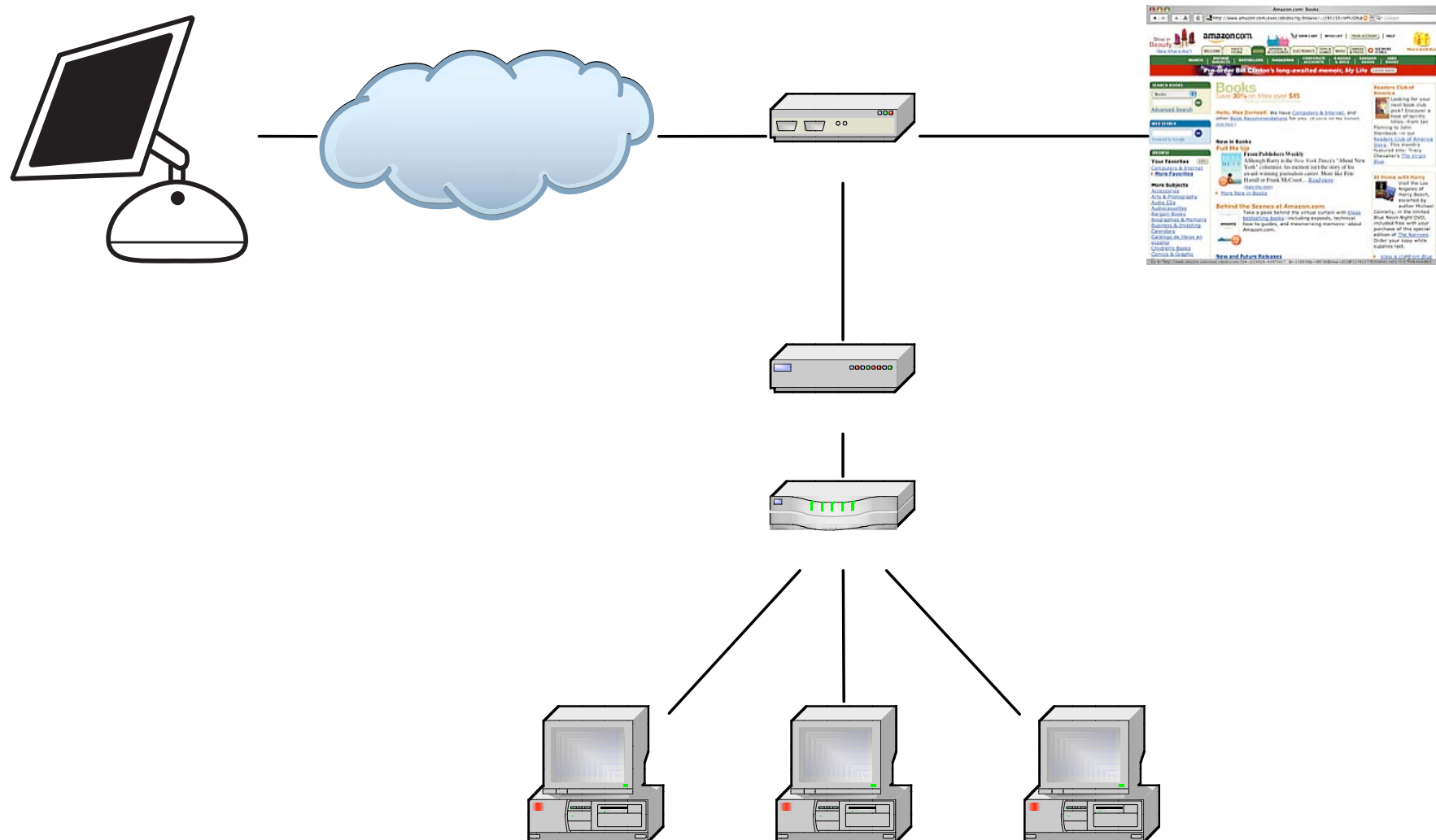
- Possible information gain on attacks and attackers.
- Possible increased security by using Honeynets as a decoy.
- Possible increased security by using aggressive Honeynets for redirection ("Bait and Switch").

Honeynet as a Decoy

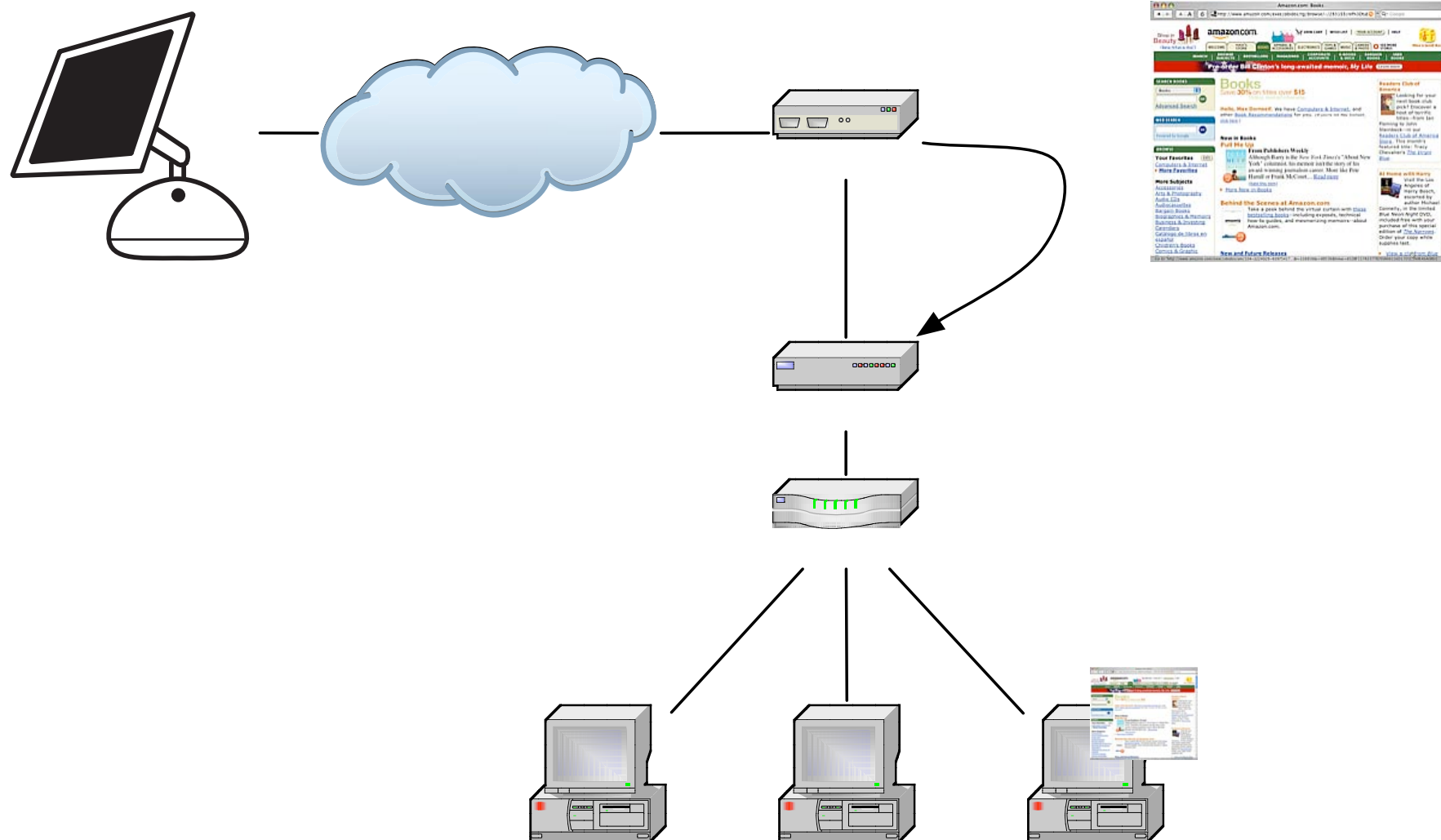


Source: Symantec Decoy Server/Mantrap Advertisement

Honeynet Redirection



Honeynet Redirection



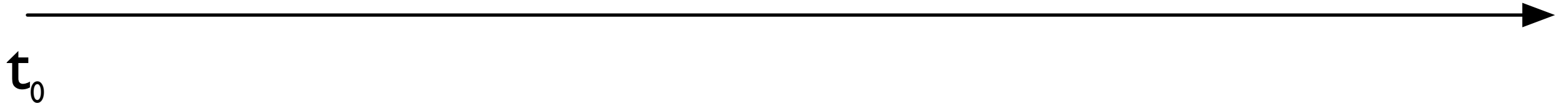
Costs of Honeynets

- Deployment
- Operation
- Increased risk
 - To the network
 - Legal liability

Building a Model

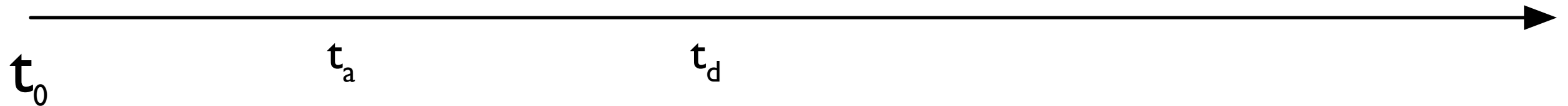
- We focus on honeypots not aimed at general security research.
 - ➡ The operator of Honeynets is not interested in all attackers but only in attacks specifically aimed at his systems. We call this type of attacker the “qualified attacker”.

Building a Model



- At t_0 the Honeynet starts operation and creates considerable fixed startup costs.
- Every unit of time the Honeynet creates cost for maintenance, housing, power, bandwidth and damages from unqualified attackers.

Building a Model



- At t_a a qualified attacker starts an attack, resulting in information gain for the operator in every unit of time.
- At t_d the attacker detects the “true nature” of the Honeynet and stops its attack.

Building a Model



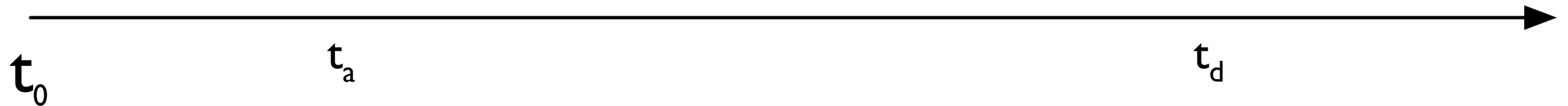
- There is no way to succeed for the attacker.
- Information gain is independent of privilege escalation.

Building a Model



- Between t_a and t_d additional costs arise.

Building a Model



- By investing in the Honeynet the Operator can make its “true nature” harder to detect thus moving t_d to the right.

Building a Model



- There is a climate of constant attack on the Internet.
- By investing in the Honeynet the frequency of qualified attacks can be increased.

Cost

$$c(t) = S + Mt$$

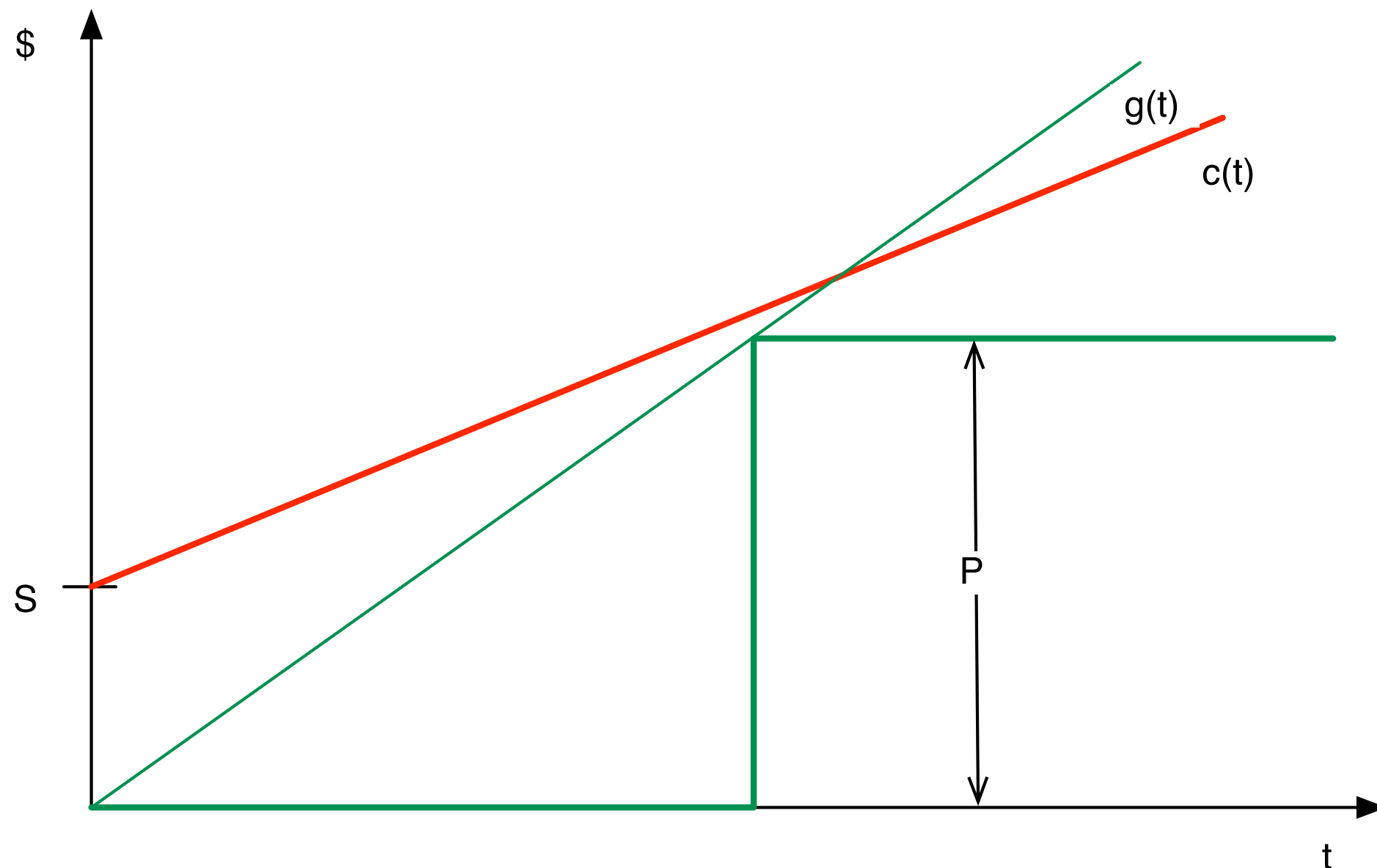
- S : start-up costs.
- M : Maintenance including forensic analysis, damage repair etc.

Utility

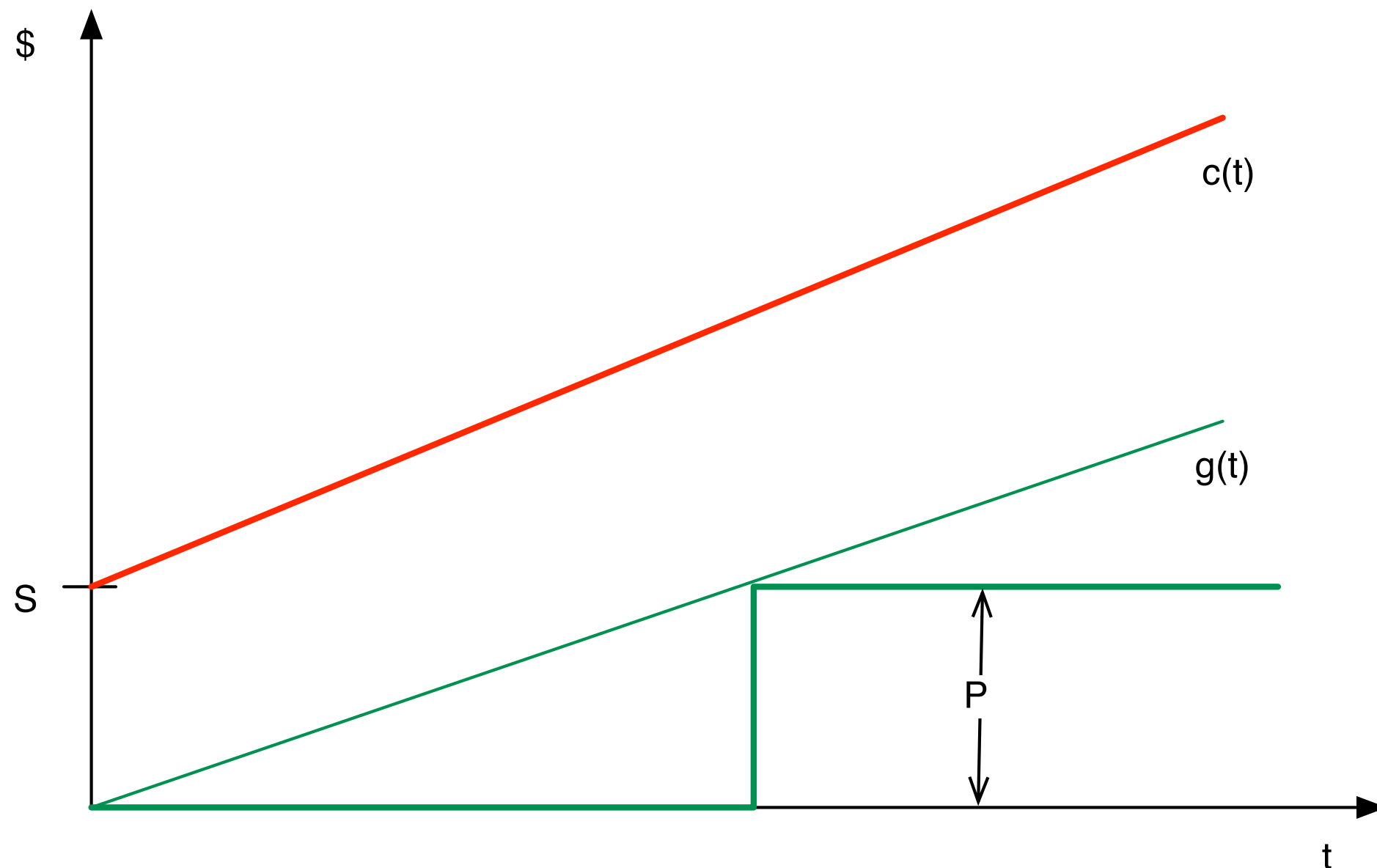
$$u(t) = Pt \frac{M}{I}$$

- M : Maintenance including forensic analysis, damage repair etc.
- P : Value of information gained by an attack.
- I : Weighting factor coupling higher investment in maintenance to increased frequency and likelihood of prolonged attack.

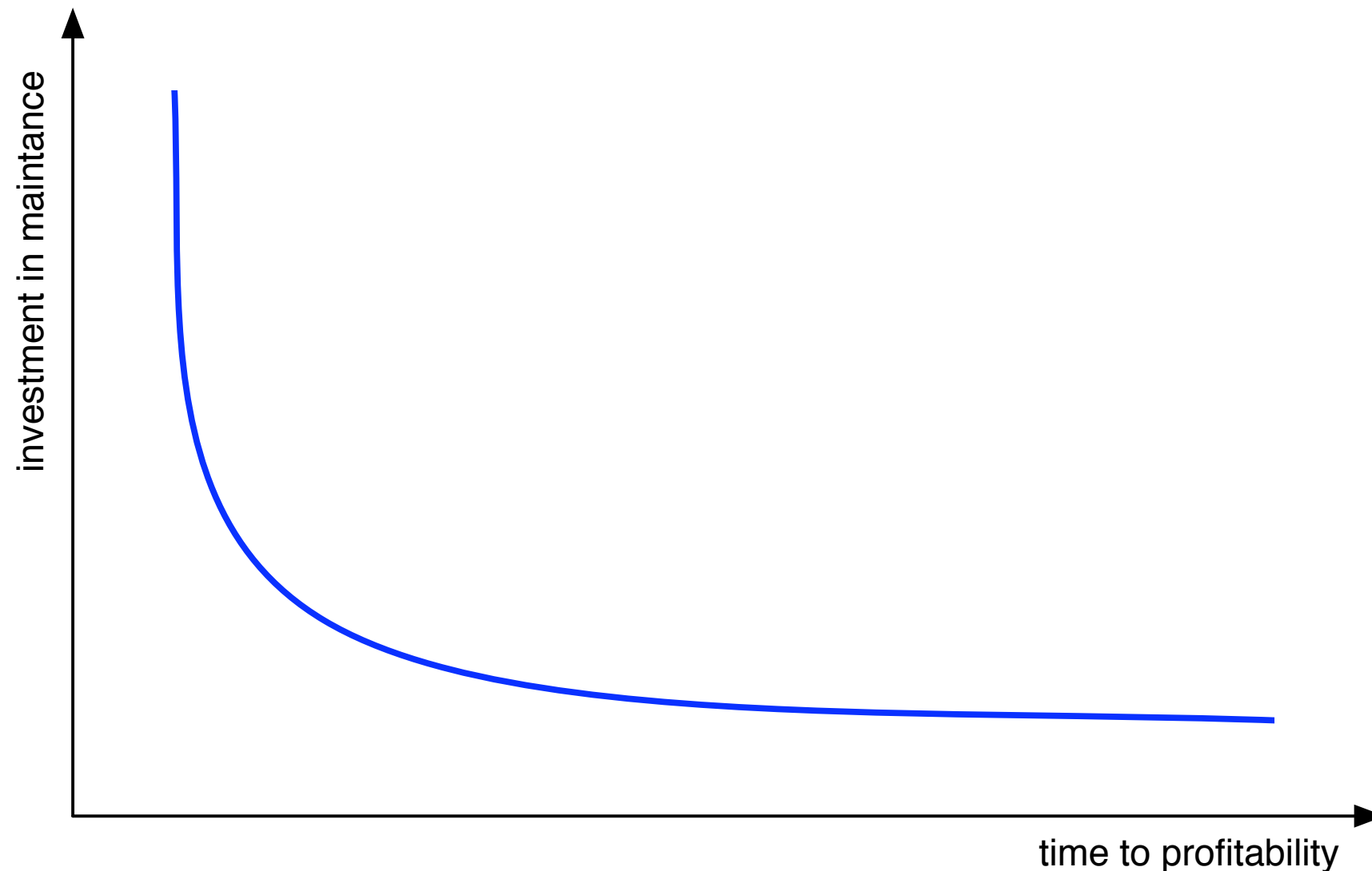
Are Honeynets profitable?



Are Honeynets profitable?



How Much Maintenance?



Real World Experiences

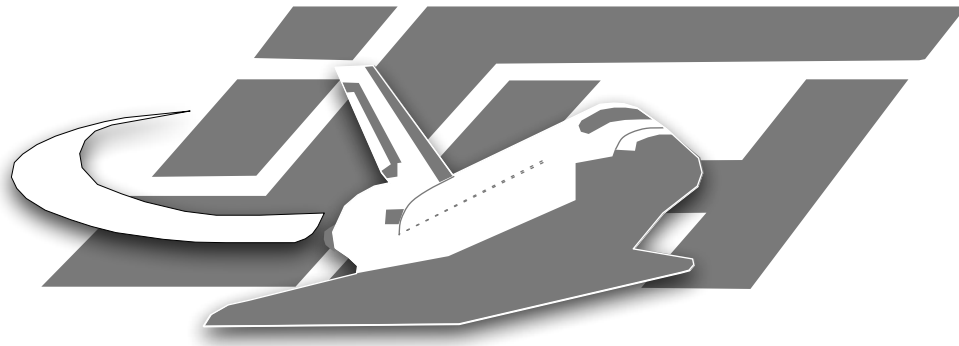
- No “qualified attackers” so far.
- Very few sophisticated attackers.
- Some formerly unknown tools captured.
- Very little learned which can’t be found out by more traditional means.
 - ➡ Except catching autonomous malware.
- Problem: redundant attacks.

Open Questions

- Can we more learn about prospective attackers?
- How and when can qualified and unqualified attackers be distinguished?
- What's about organizations interested in all attackers?
- What is the value of the information gained by a qualified attack (I)?

Open Questions

- How does more investment in maintenance (decoy) relate to more and longer attacks?
- Is the gain by an qualified attack linear with duration of attack?
- Where can results obtained by Honeynet obtained by other possibly cheaper means?
- Get the Numbers!



The costs and benefits of Honeynets

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Slides at
<http://md.hudora.de/presentations/#weis2004>